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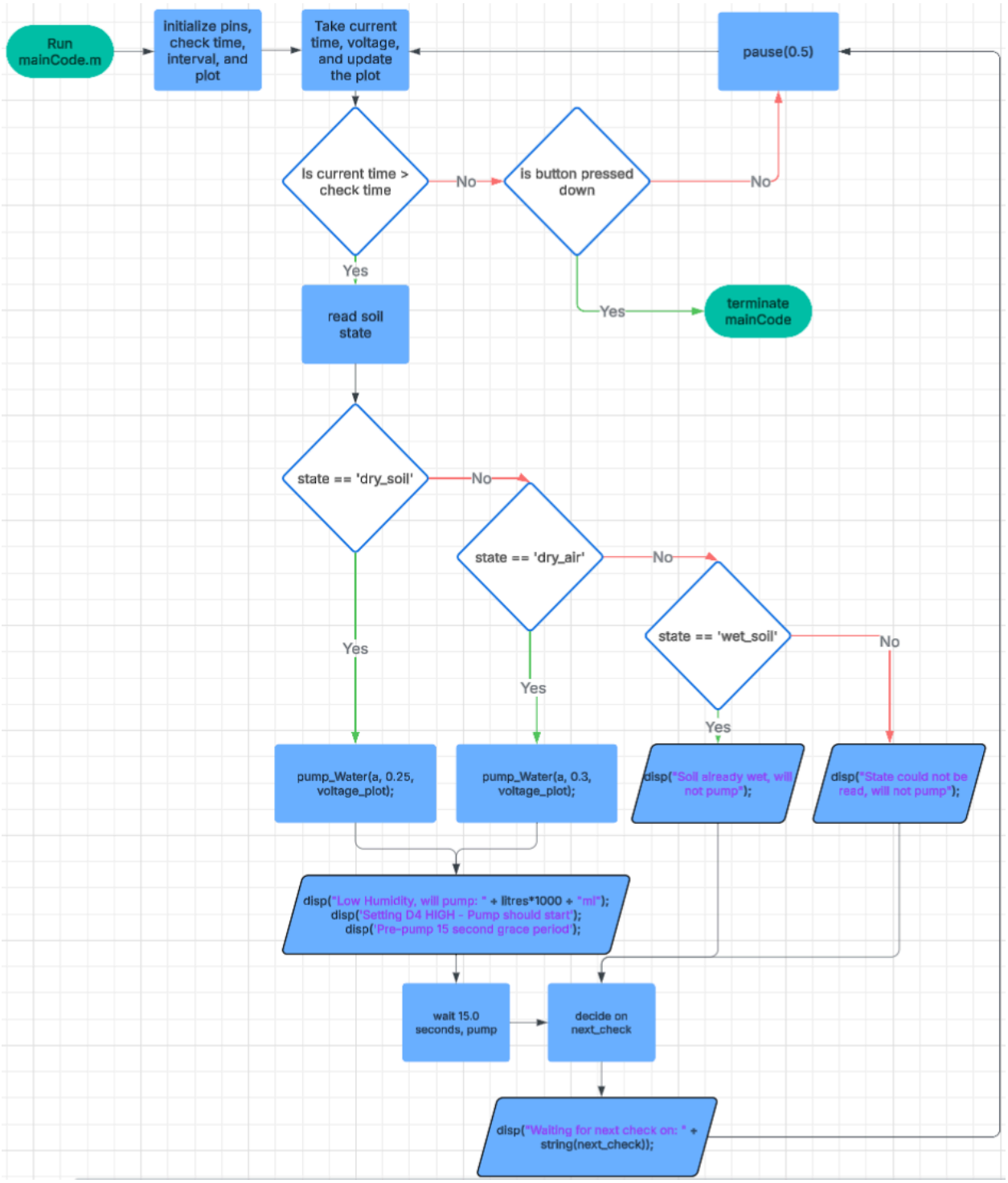
April 10th 2025

Automatic Water Pump Project

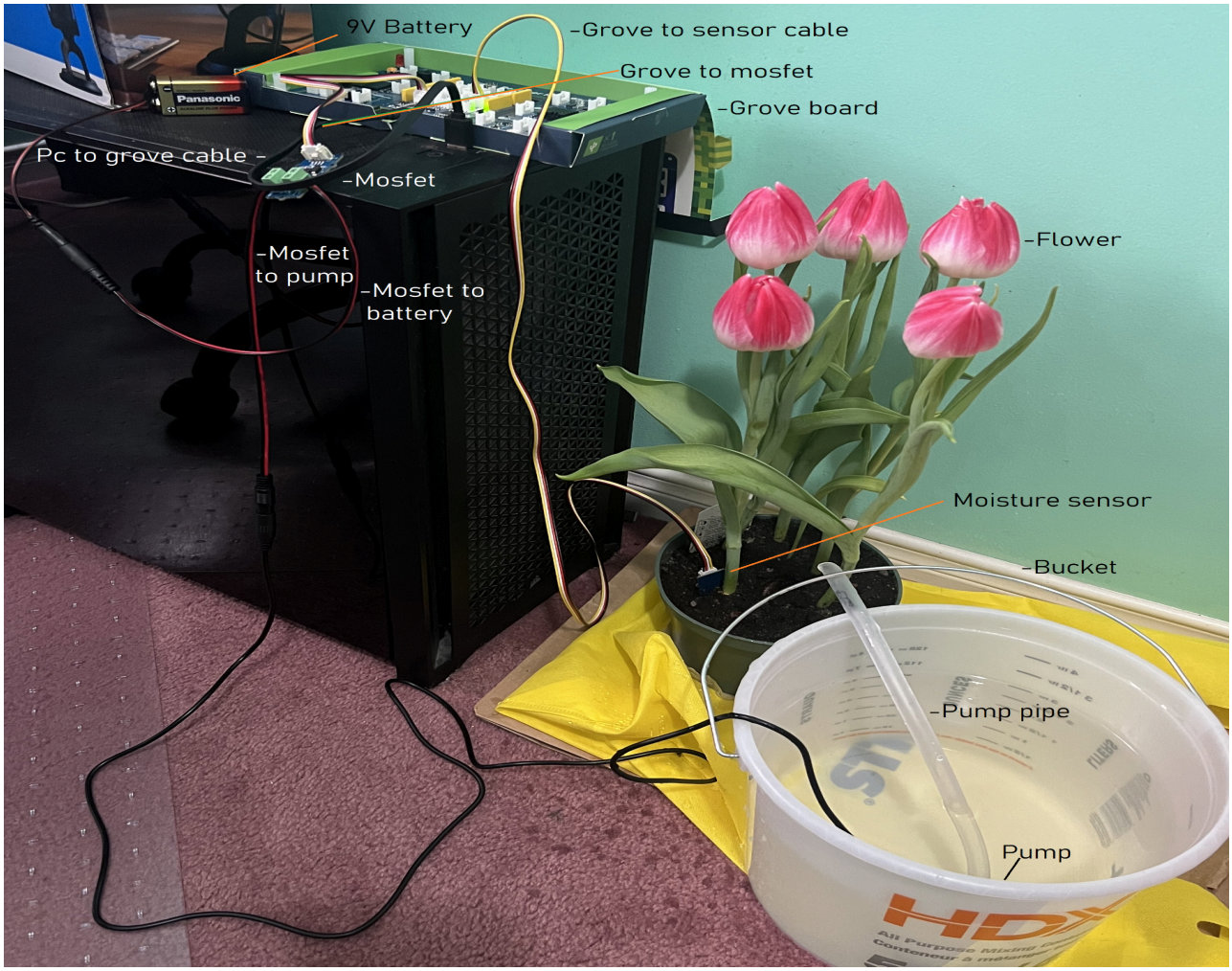
This project focuses on designing and creating an automatic water pump. This project aims to create a robot that dispenses water to a potted plant when certain conditions are met. To meet what potted plants usually require, the soil's dry and wet moisture levels were read and stored. Water will be dispensed if enough time has passed and the soil is dry. MATLAB was used to read moisture, using a soil moisture sensor, and calculate how much time has passed. The robot successfully reads the moisture every 0.5 seconds and dispenses water if a set amount of time has passed. This code assumes the water need is 250ml or 300ml per pump cycle. While the checks can be done often, it is recommended to add other options for future iterations.

The soil moisture levels were read and used to establish voltage thresholds. Through sensor calibration, I have arrived at these values: [>2.2V = Dry Air], [1.85- 2.2V = Dry Soil], [<1.85V = Wet Soil]. Pump activation duration is based on its rate of 1 liter per 24 seconds.

This part aims to visualize the robot's steps in the shape of a flowchart. It is included on the next page and was made using the Flowchart template of Lucid.app, following ICT diagram conventions. The following logic ensures that water is dispensed only when both time and moisture conditions are met, and the code can be terminated via a button press.

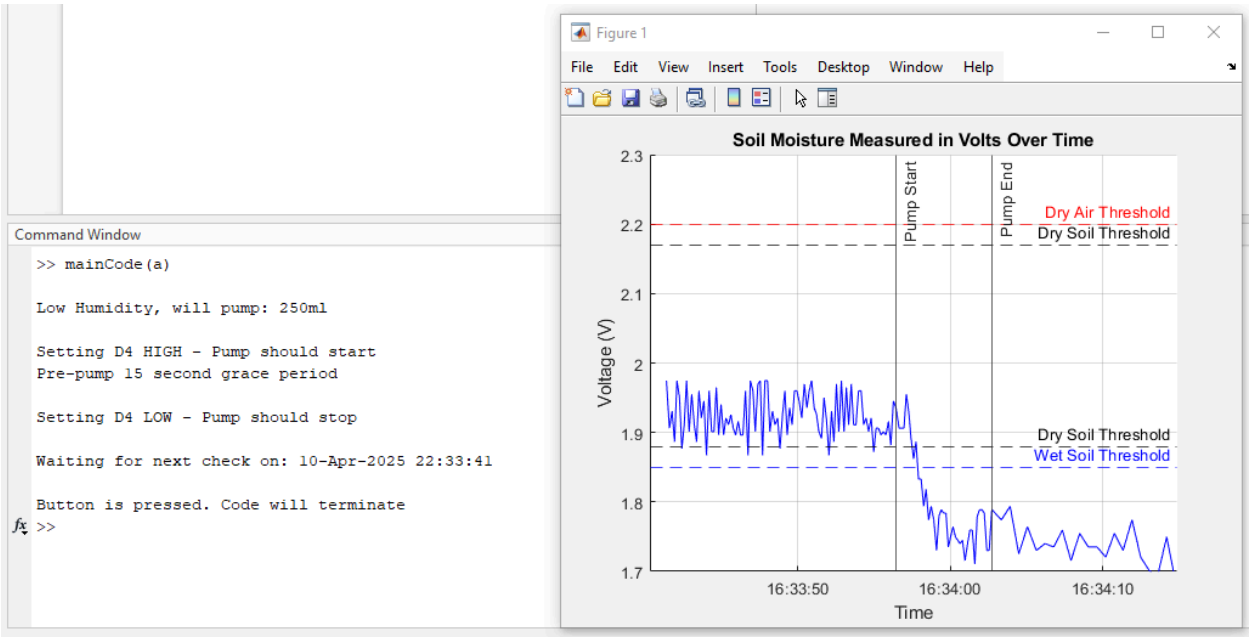


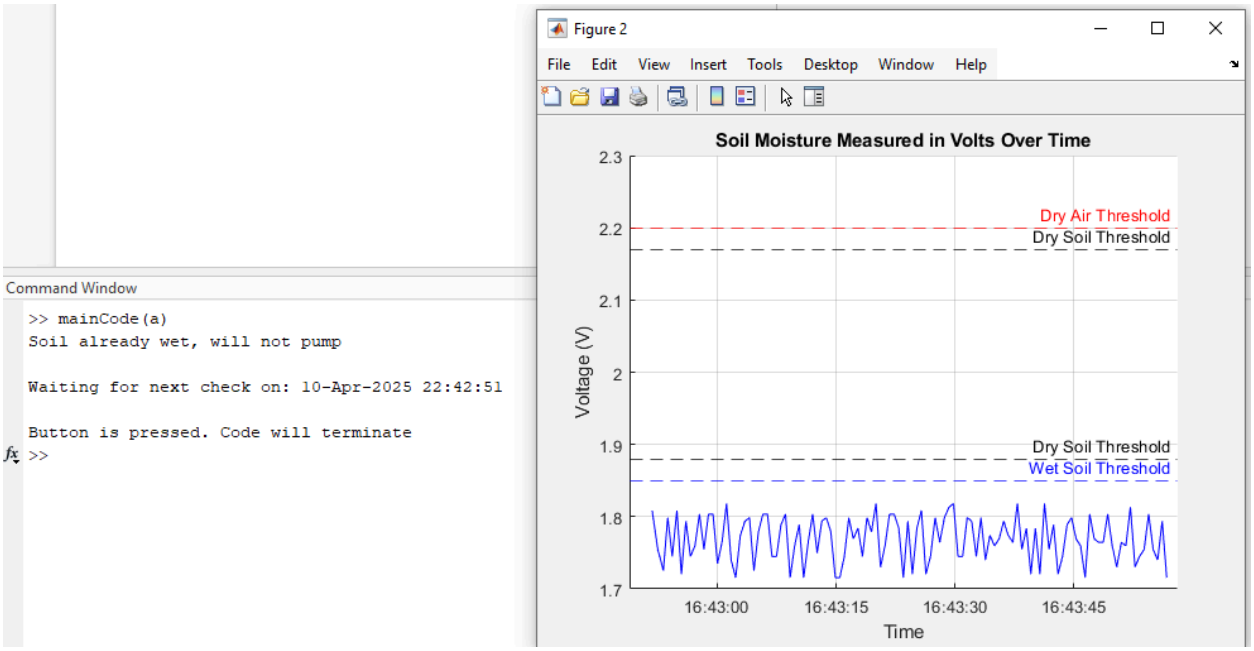
The upcoming image will include every part needed for the machine, along with its name and use. Additionally, the code will utilize Grove's onboard LED to turn on while the pump is working, and the button will be used as a "terminate code."



First of all, a Grove board. This will allow us to utilize the sensor and use the pump with its pin interface. Any USB cable that can be inserted into the Grove will be sufficient. Grove cables to utilize the moisture sensor and the pump by connecting them to the Grove. The moisture sensor will allow us to read, log, and plot the moisture readings of the soil. Mosfet, power cables to the pump, and the battery. This will allow us to send commands to the MOSFET, making the battery turn the pump on. The pump and the battery will be connected to the left and right power cables in that order. Lastly, a potted flower and a container for the water.

Outputs of the code before and after the initial run of the code:





The slight differences between the thresholds at both ends are simply for visual clarity. The exact thresholds are at 1.85 and 2.2